

Time Series Analysis on Disney's Animal Kingdom Wait Times

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Abstract—This study attempts to extract the seasonal nature of park attendance and its intricacies within each day using STL.

I. INTRODUCTION TO DATASET

The data set is composed of the wait times in minutes of every ride in Animal Kingdom with a frequency of around 10 min from May 26, 2017 to February 24, 2019. For STL we padded -999 with the previous wait time and removed closed hours. For STAHL we replaced them with NaN alongside the park closed hours.

II. METHODS

Time Series Decomposition

To unravel the underlying patterns within the Animal Kingdom's wait times, this study utilizes Seasonal and Trend decomposition using Loess (STL). STL is a very flexible and robust method used to decompose time series data. The algorithm makes the assumption of an additive decomposition model, showcasing the observed time series y_t as the sum of three distinct components:

$$y_t = T_t + S_t + R_t$$

- **Detrending:** The algorithm temporarily removes an initial trend estimate to isolate the seasonal patterns.
- **Seasonal Extraction S_t :** LOESS smoothing is applied to the detrended series to estimate the seasonal component for each specific period.
- **Trend Extraction T_t :** The newly estimated seasonal component is subtracted from the original series. The remainder is then smoothed again using LOESS to capture the underlying long-term trend.
- **Remainder Calculation R_t :** The residual component is calculated by subtracting both the finalized trend and seasonal components from the original observed series.

The core mechanism of the STL algorithm and what differentiates it from its classical rigid method is Locally Estimated Scatterplot Smoothing (LOESS). For comparison your classical method fits a global function across the entire dataset, LOESS is a non-parametric regression method that fits simpler models to localized subsets of the data. This localized approach allows the model to smoothly connect data points

based on their immediate neighbor helping deal with complex data sets. Hence, STL was picked for our data set due to the complex seasonal cycles. [1]

III. RESULTS AND DISCUSSION

Figure 1 poses a challenge to interpret, thus having Figure 2 with its 60-day window. Figure 2 shows STL extracting the opening and closing nature of a theme park in its seasonal component. Alongside Figure 3, it tells us more of the story; the seasonal component extracted the daily ups and downs of park attendance. In other words, when the park opens/closes, the wait times go up/down. Figure 4 showcases the three busiest days of the year. We first have our trend that gradually goes down as the year comes to a close. For each day, we can see the morning rush, the lunch dips, the afternoon rush, and the evening dip. Our residual term is in much better showing no patterns for 2/3 days. So, STAHL is a different type of decomposition that attempts to account for a holiday component. After some downsampling of the data, technically, it did extract our component in Figure 5. Figure 6 shows the challenge of real-world data. Since we replaced our values with NaN, the model struggled to find the patterns of our data. [2]

IV. CONCLUSION

The seasonal component for 60 days is the highlight of this study we were able to extract the ups and down from theme park attendance in our seasonal components. Furthermore, on three busiest days, the seasonal component was able to extract the rushes and lower times of theme park attendance.

V. REFERENCES

REFERENCES

- [1] R. B. Cleveland, W. S. Cleveland, J. E. McRae, and I. Terpenning, "Stl: A seasonal-trend decomposition procedure based on loess," *Journal of Official Statistics*, vol. 6, no. 1, pp. 3–73, 1990.
- [2] D. Haller and G. Bellone, "Stahl: Seasonal, trend, and holiday decomposition with loess," *Journal of Official Statistics*, vol. 41, no. 4, pp. 1112–1153, 2025.

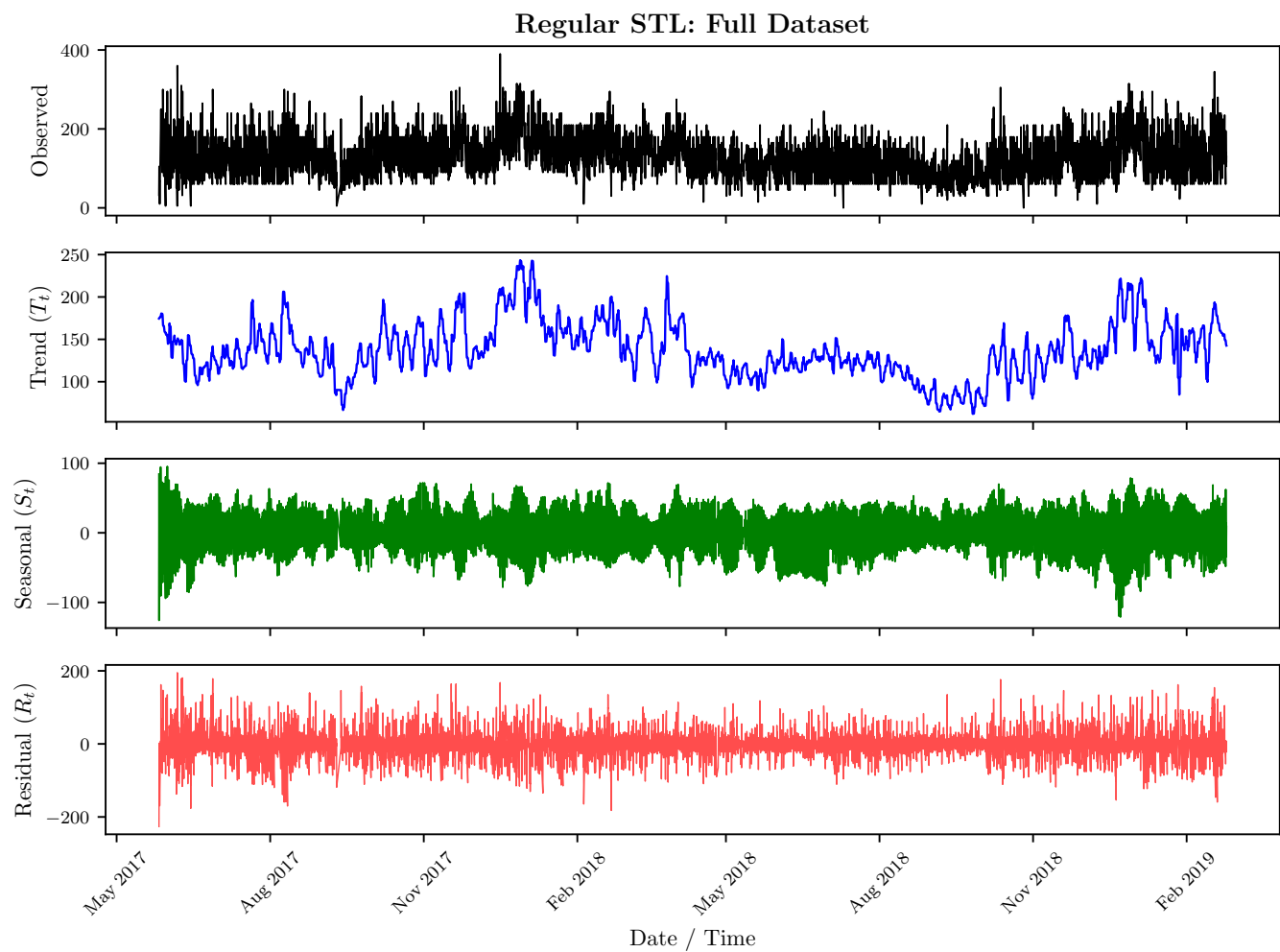


Fig. 1. STL decomposition of the wait-time dataset. The panels illustrate the original observed signal alongside its decomposed components: the long-term trend, the periodic seasonal patterns, and the stochastic residual (remainder). The x-axis represents the temporal progression across the full study period, while the y-axis denotes wait times in minutes.

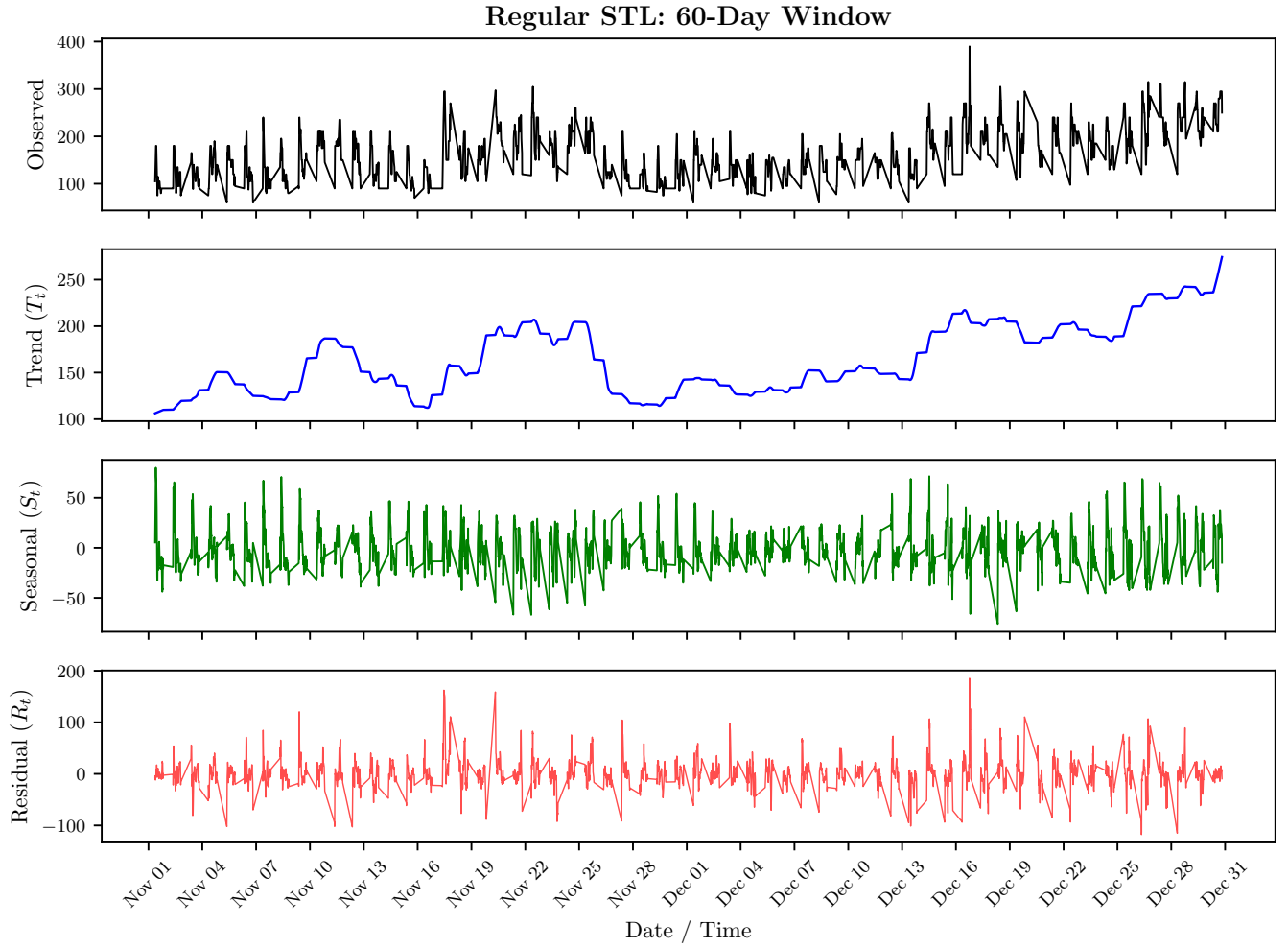


Fig. 2. STL decomposition on a 60 day window in 2017 on AK wait time dataset. The panels illustrate the original observed signal alongside its decomposed components: the long-term trend, the periodic seasonal patterns, and the stochastic residual (remainder). The x-axis represents the temporal progression across the full study period, while the y-axis denotes wait times in minutes.

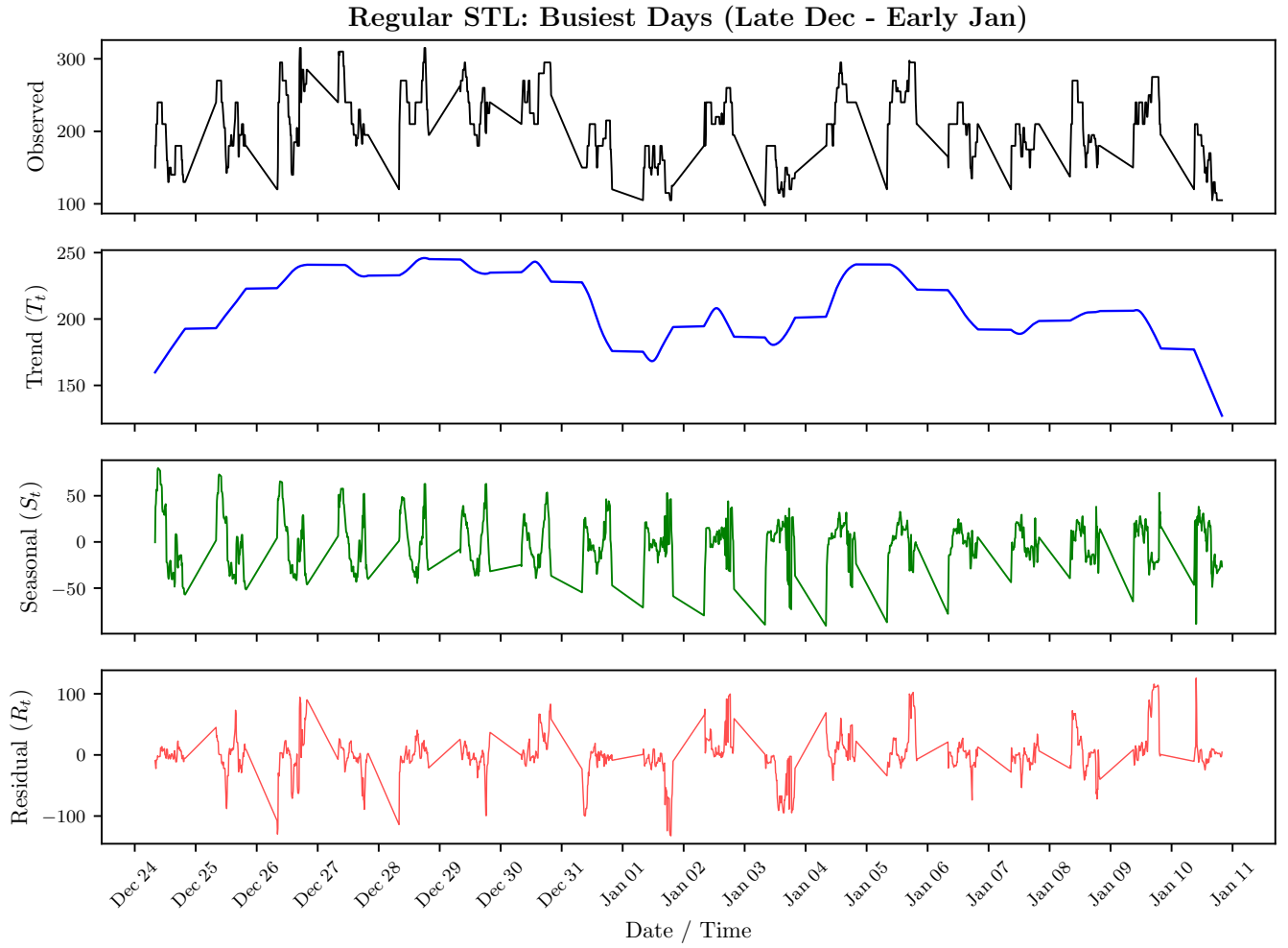


Fig. 3. STL decomposition on a busiest days in 2017 to 2018 on AK wait time dataset. The panels illustrate the original observed signal alongside its decomposed components: the long-term trend, the periodic seasonal patterns, and the stochastic residual (remainder). The x-axis represents the temporal progression across the full study period, while the y-axis denotes wait times in minutes.

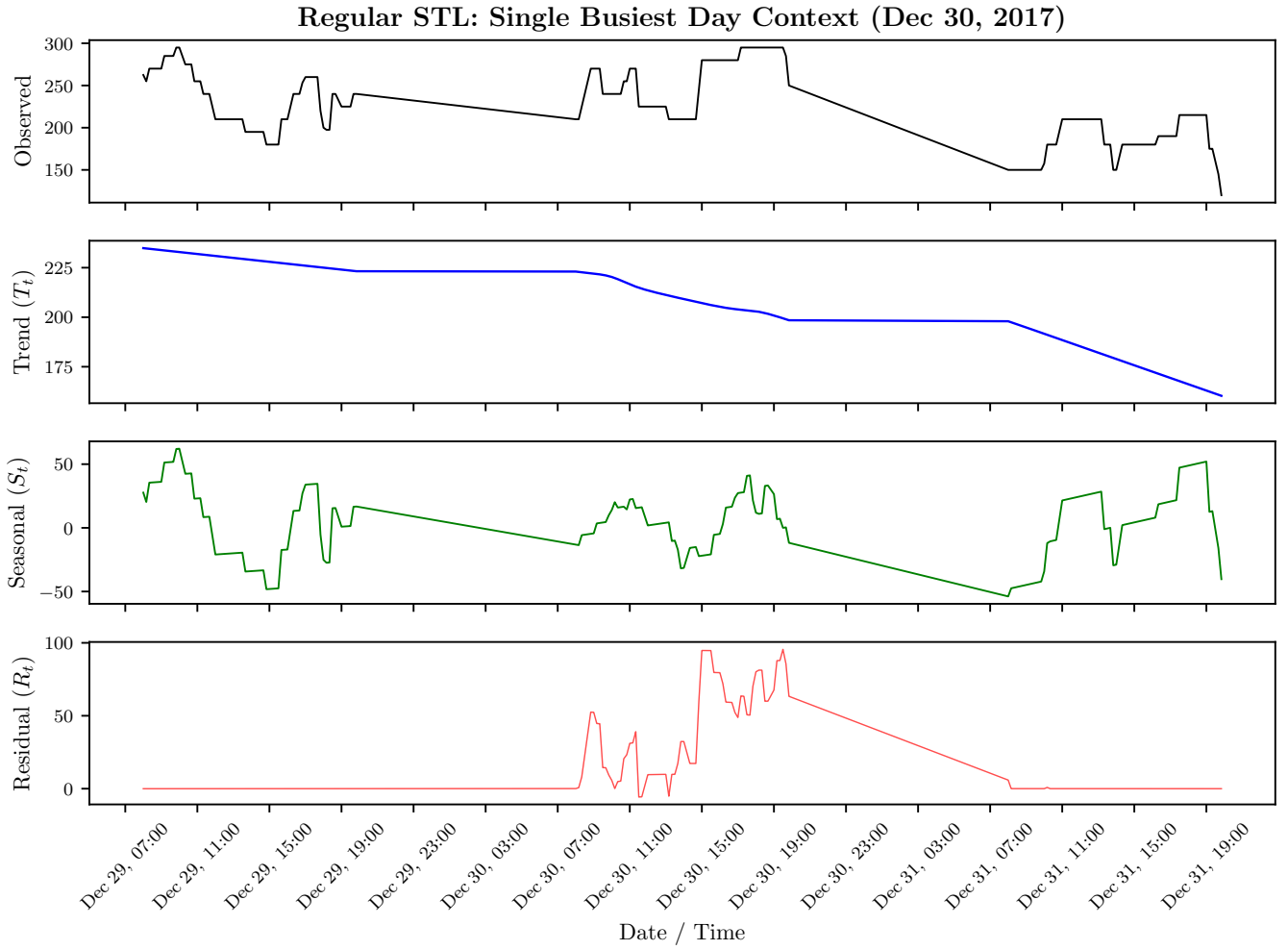


Fig. 4. STL decomposition focusing on the single busiest day (Dec 30, 2017) using a 3-day context window. The panels illustrate the original observed signal alongside its decomposed components: the long-term trend, the periodic seasonal patterns, and the stochastic residual (remainder). The x-axis represents the hourly temporal progression, while the y-axis denotes wait times in minutes.

STAHL: Tracking the Busiest Day of the Year

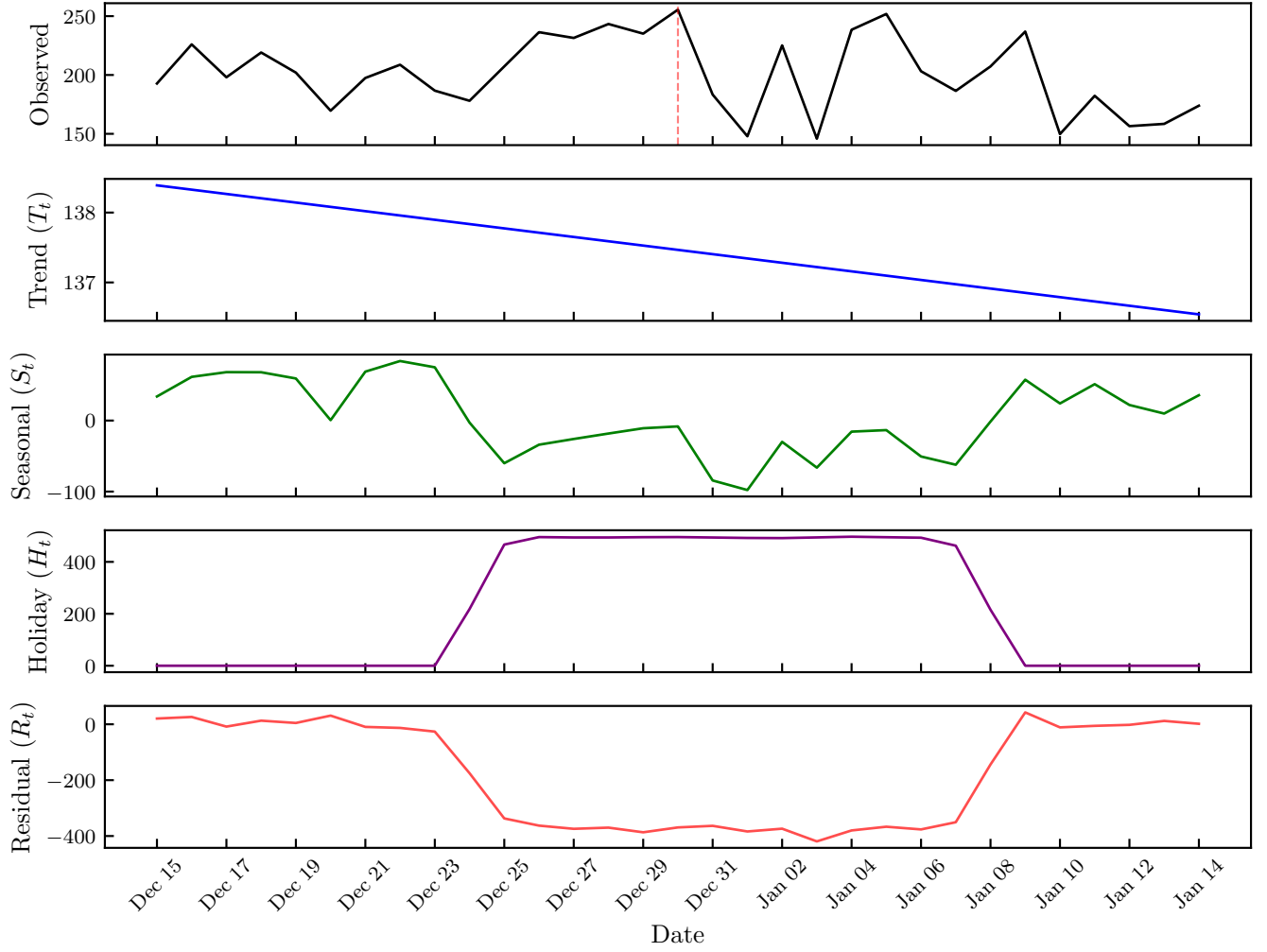


Fig. 5. STAHL decomposition zoom (incorporating downsampling) on the busiest 3 days in 2017 to 2018 on AK wait time dataset. The panels illustrate the original observed signal alongside its decomposed components: the long-term trend, the periodic seasonal patterns, and the stochastic residual (remainder). The x-axis represents the temporal progression across the full study period, while the y-axis denotes wait times in minutes.

STAHL: High-Res 5-Min Wait Times (Top 3 Busiest Days)

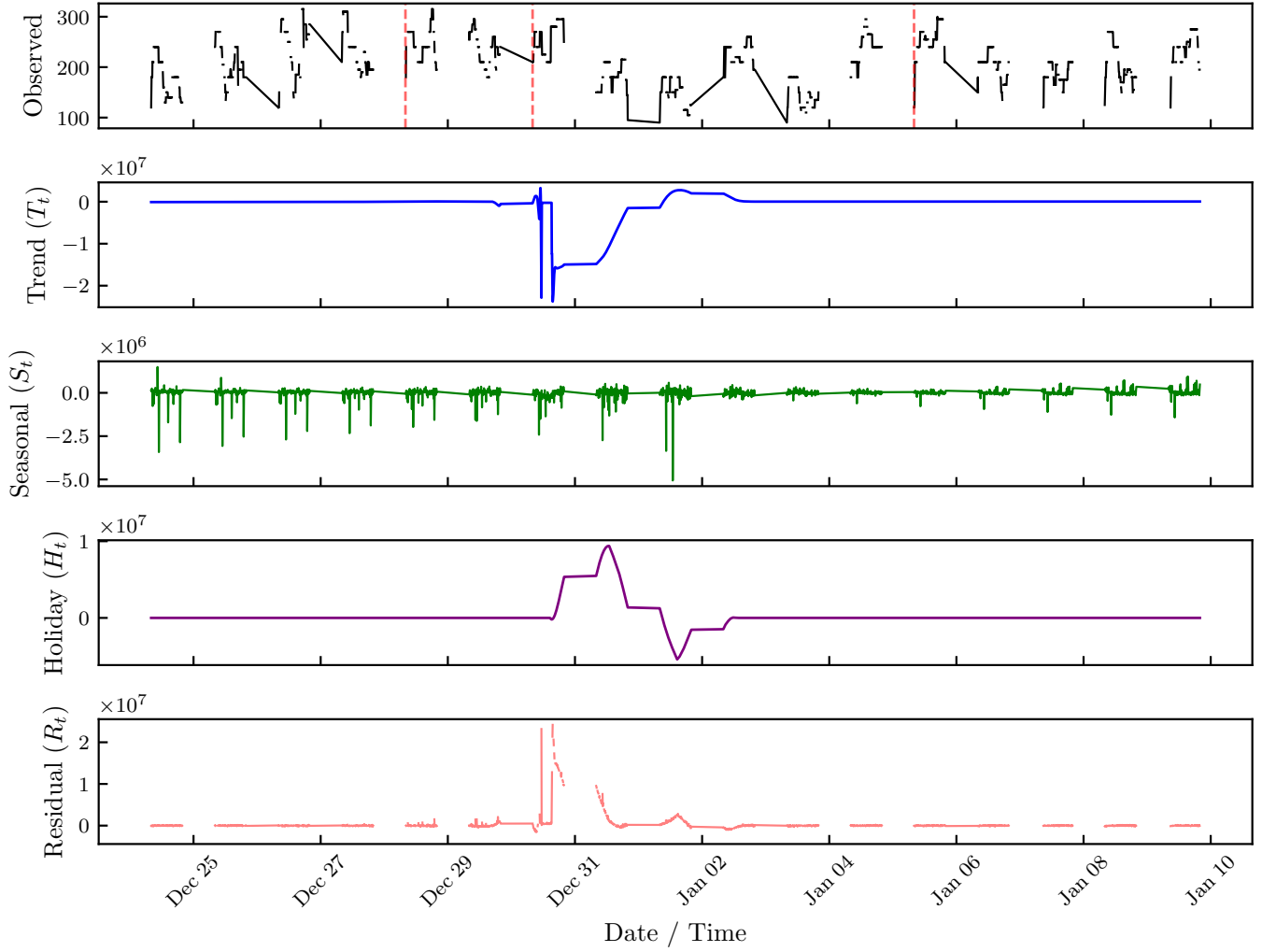


Fig. 6. STAHL high-resolution (5-minute interval) decomposition on the busiest 3 days in 2017 to 2018 on AK wait time dataset. The panels illustrate the original observed signal alongside its decomposed components: the long-term trend, the periodic seasonal patterns, and the stochastic residual (remainder). The x-axis represents the temporal progression across the full study period, while the y-axis denotes wait times in minutes.